

Product Sentiment Analyzer and Review Dashboard

PROJECT DESCRIPTION:

The **Product Sentiment Analyzer and Review Dashboard** is a web-based application that collects product reviews from e-commerce websites such as Amazon and Flipkart using dynamic web scraping. The application analyzes the reviews using sentiment analysis techniques and displays the insights on an interactive dashboard.

Reviews are classified as positive, negative, or neutral using NLP libraries. Users can search for a product, view its reviews, and explore visual sentiment trends through interactive charts. The scraped and processed data is stored in a database and presented via a user-friendly interface. The application is hosted using free cloud services for broader accessibility.

FEATURES:

- 1. Dynamic Review Scraping**
Scrapes reviews in real-time from major e-commerce sites using Selenium.
- 2. Sentiment Classification**
Analyzes the sentiment of each review using TextBlob(NLP) or VADER.
- 3. User Interface for Search and Visualization**
Enables users to search for products and view sentiment charts and reviews.
- 4. API Integration**
Backend APIs serve data to the frontend and handle user queries securely.
- 5. Database Connectivity**
Stores product data, reviews, and sentiment analysis results in MongoDB or PostgreSQL.
- 6. Visual Analytics**
Displays sentiment distribution, trends over time, and word frequency using charts.
- 7. Cloud Hosting**
Deployed using platforms like AWS EC2, Render (for backend), and Vercel/Netlify (for interface).

TECHNOLOGIES USED:

Interface Technologies:

- React.js
- Chart.js or Recharts
- Axios for API communication

Backend Technologies:

- Flask or Django
- Web scraping with Selenium and BeautifulSoup

- Sentiment analysis using TextBlob or VADER
- pandas and NumPy for data handling

Database:

- MongoDB Atlas or PostgreSQL (via ElephantSQL or Supabase)

Deployment Platforms:

- Render or AWS EC2 (for backend)
- Netlify or Vercel (for interface)
- MongoDB Atlas (for database)

OUTCOMES:

1. **Product Review Insights**
Helps users understand the general sentiment around a product before purchase.
2. **Business Intelligence**
Enables companies to identify strengths and weaknesses based on customer feedback.
3. **Data-Driven Reports**
Provides visual data insights such as rating trends and frequently used terms.
4. **Review Monitoring**
Offers real-time monitoring and historical tracking of product sentiment.